

Print Data Loss Prevention (Print DLP)

Table of contents

1 Summary..... 3

2 Configure Print DLP..... 4

3 Configure direct printing with PDLP 8

3.1 Turn off spool file pooling.....10

4 Configure pull printing with PDLP..... 13

4.1 Turn off spool file pooling.....16

1 Summary

Print Data Loss Prevention (Print DLP) is a feature within HP Access Control's Intelligent Print Management (IPM) to apply rules-based printing based on document content. This allows for restriction of printing if a document contains specific keywords or patterns. For example, any document containing the word "confidential" or the numeric pattern of a social security number can be prevented from printing.

IMPORTANT: Postscript print drivers are currently the only supported drivers.

NOTE: Keywords or patterns as a watermark within documents is not currently supported.

NOTE: Text within images is not currently supported.

To install Print DLP for use with point-and-print printers, select the Print DLP bundle during the installation of HP AC. To use Print DLP with pull printing, install the HP AC Enterprise bundle.

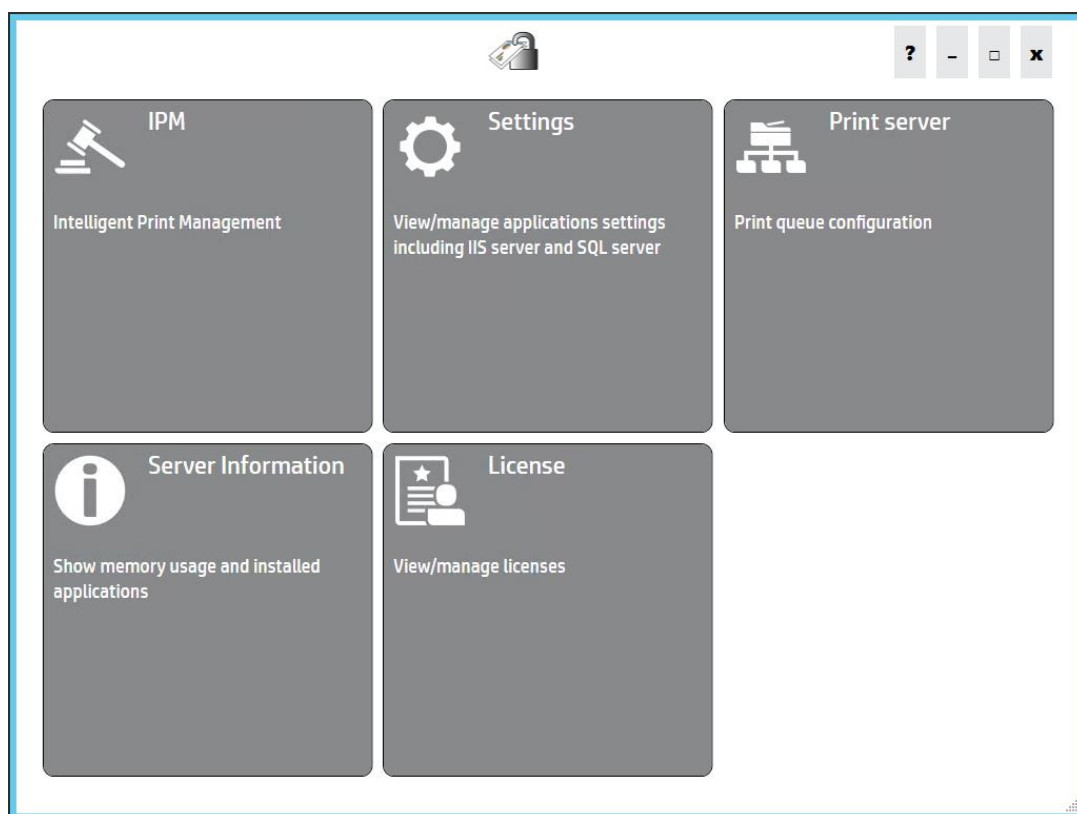
The Print DLP license is part of IPM, and is temporarily licensed for 30 days. To verify that Print DLP is licensed, go to the HP AC admin console > License tile, and verify "Print DLP" displays in the HP AC IPM license section.

The screenshot shows the 'License' section of the HP AC admin console. It features a 'Server' field with the value 'EOMCTQA01' and an 'Upload license' button. To the right, a text block provides instructions on how to contact HP licensing support. Below this, the 'HP AC IPM' section displays the 'Expiration date' as '10/20/2017' and '0/10 device(s)' licensed. The 'Print DLP' label is visible at the bottom right of the IPM section.

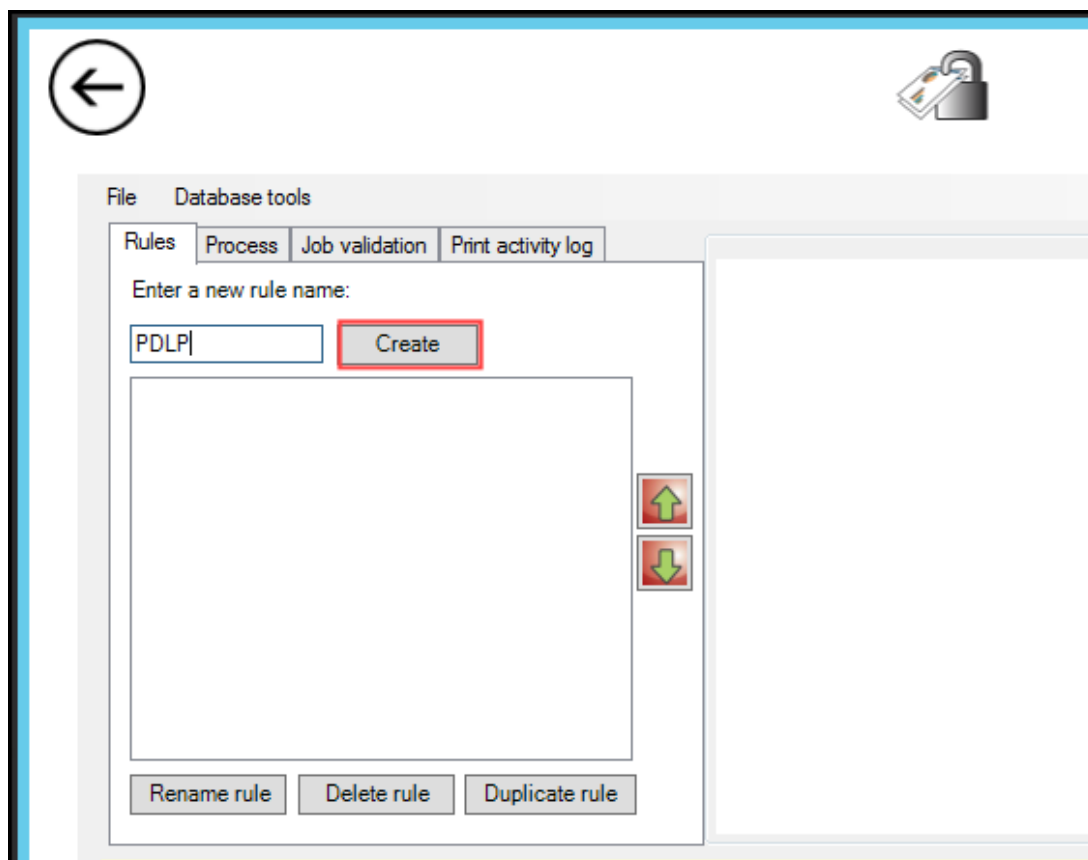
License	
Server	EOMCTQA01
 Upload license	
To contact a licensing customer support center, go to http://licensing.hp.com and click on the "contact HP licensing" link on the left. The form provided allows you to send requests to the licensing support team.	
HP AC IPM	
Expiration date	10/20/2017
	0/10 device(s)
Print DLP	

2 Configure Print DLP

1. On the HP AC home screen, select the IPM tile.



2. In the Rules tab, enter a name for the new Print DLP rule, then click Create.



3. In the Process tab, click and drag the Job condition from the Conditions list to the process window. Then right-click the condition and click Properties.



4. In the Job properties window, select Job Content, then select Contains, then enter the words or patterns to search for in documents. When finished, click OK.

The screenshot shows a window titled "Job" with a close button (X) in the top right corner. Below the title bar, it says "Apply the following condition when :". On the left, there is a list of conditions with checkboxes: "Job name", "Number of pages", "Job size (bytes)", "Color job", "Application name", and "Job content". The "Job content" checkbox is checked. To the right of this list is a smaller list with a checkbox and the word "Contains", which is also checked. To the right of this, there are five text input fields. The first field contains the word "Confidential". The second field contains the regex pattern "^d{3}-d{2}-d{4}\$". The third field contains the words "Top Secret". The fourth and fifth fields are empty. At the bottom left of the window is an "Ok" button, and at the bottom right is a "Cancel" button.

NOTE: There are 5 fields for each Job condition. If any one of the fields are found in the document, the rule will be triggered.

NOTE: To search for a pattern, use Regex patterns. Regex patterns, such as social security numbers or phone numbers, can be found online (<http://regexlib.com/Search.aspx>). The Regex pattern in the image above is for a social security number.

5. Click and drag an action from the Actions list to the process window, such as the "Do not print" action. This means that when the job content is found in a document, the document will not print.



6. Click and drag a notification from the Notifications list to the process window, such as the Client Application notification. This requires the IPM Notification Desktop Client to be installed on user workstations and allow for a pop-up to notify the user if the IPM rule was followed. In the process window, right-click the Client Application notification icon and click Properties to customize the pop-up message.

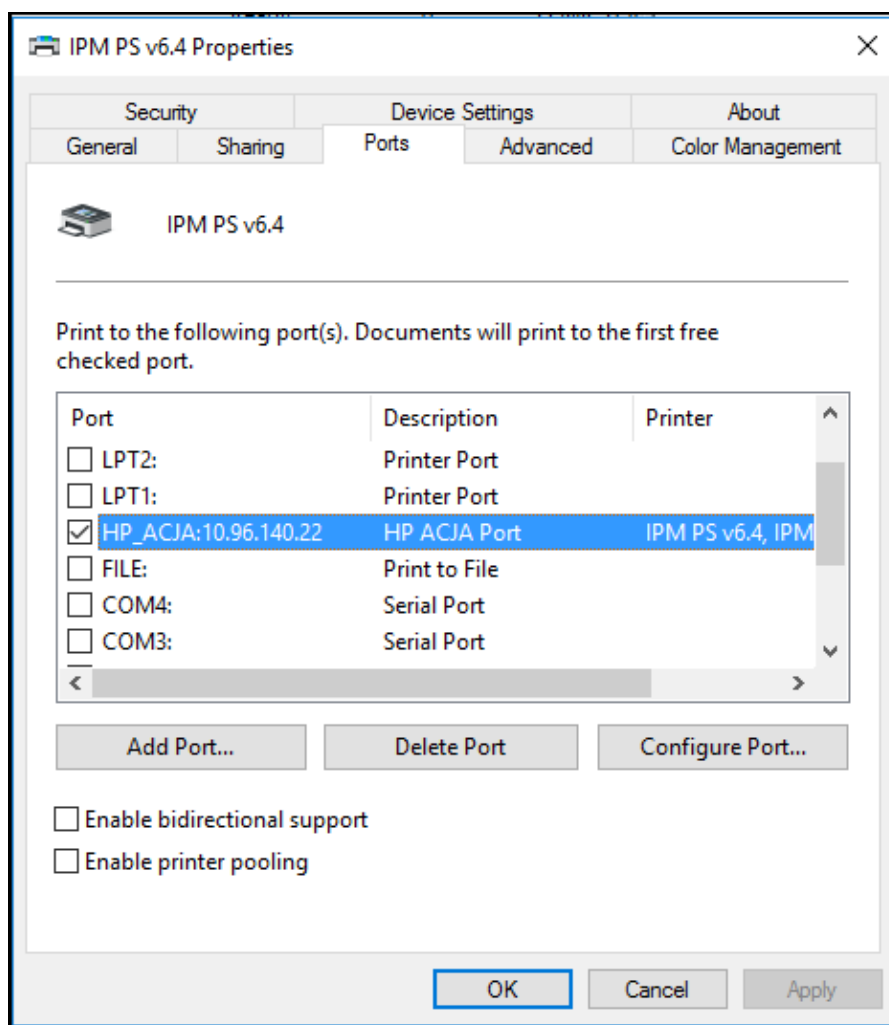


3 Configure direct printing with PDLP

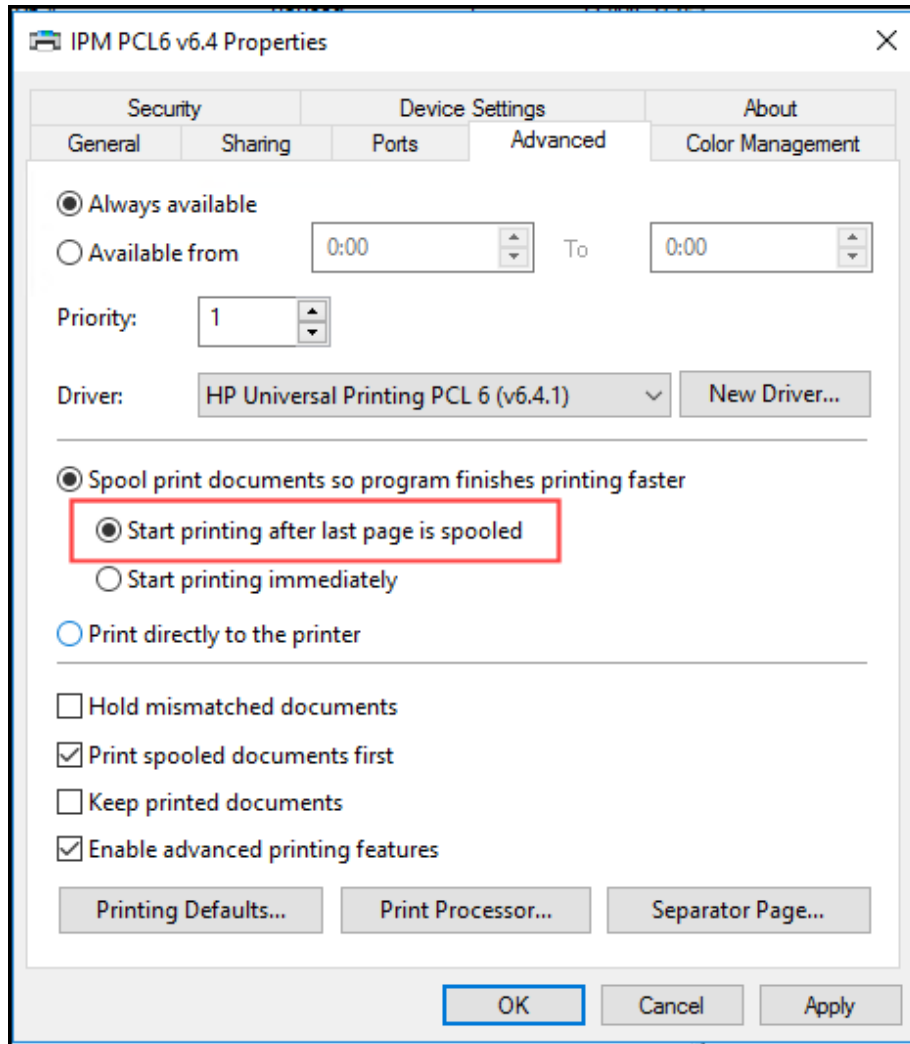
1. On the HP AC server, add a direct IP print queue and be sure to enable this printer for sharing.
2. Go to HP AC > Print Server and configure the printer for IPM.

	Name 	Port 	IPM 
	P3015	10.96.140.44	☐
	Microsoft XPS Document Writer	PORTPROMPT:	☐
	Microsoft Print to PDF	PORTPROMPT:	☐
	IPM PS v6.4	10.96.140.75	☒
	Express	exd	☐

3. Verify the printer is correctly configured by going to the printer properties > Ports tab, and making sure the port begins with "HP ACJA:".



4. In the printer properties > Advanced tab, make sure "Start printing after last page is spooled" is checked.



5. From user workstations, connect to the shared printer. When users print any document to this shared printer, the IPM rule will be checked.

3.1 Turn off spool file pooling

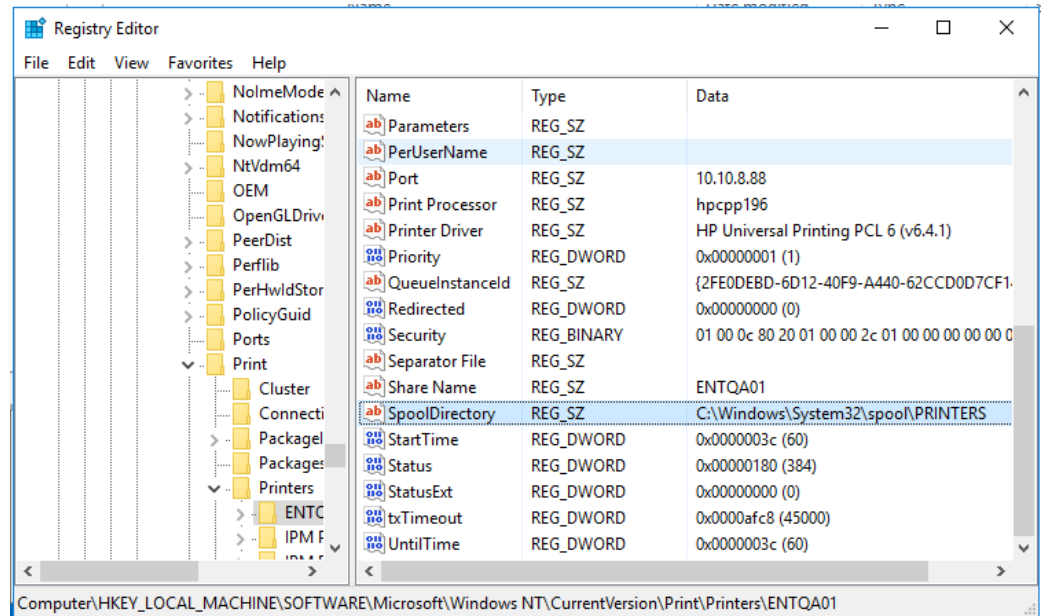
For PDLP to read document content, spool file pooling must be turned off to store spool files as SPL file types. Follow the steps below to turn off spool file pooling per print queue or for all printers:

- To turn off spool file pooling per print queue:
 1. On the HPAC server, open the Registry Editor.
 2. Locate the following registry subkey, where <printername> is the name of your printer:

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Print\Printers\<printername>

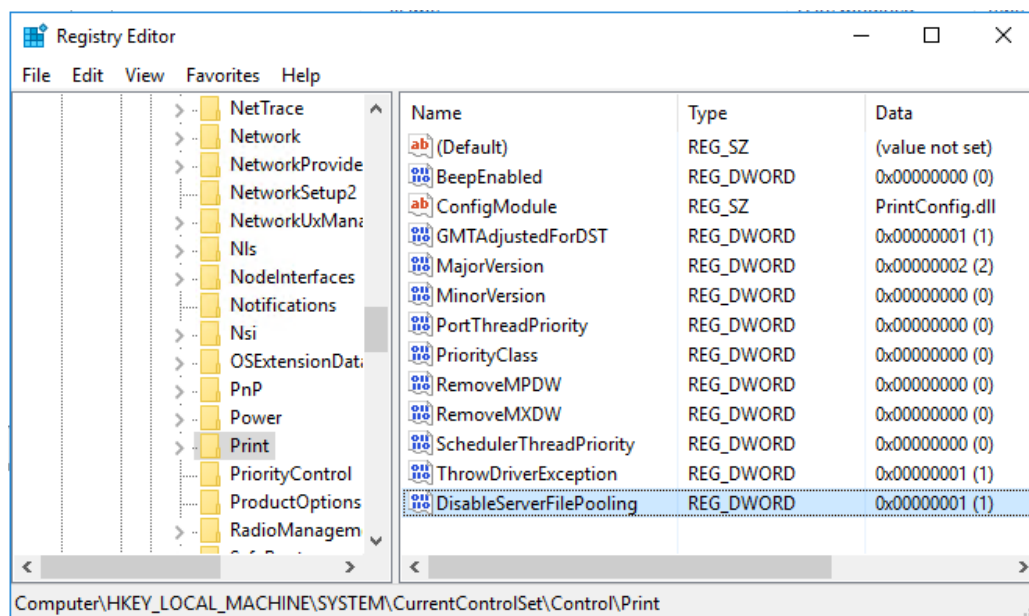
3. In the right pane, right-click SpoolDirectory, and then click Modify.
4. In the Value data box, type the location where you want the spool files to be stored. For example, type C:\Windows\System32\spool\PRINTERS as the default location.

IMPORTANT: The “\PRINTERS\<printername>\SpoolDirectory” location must be the same as the “\PRINTERS\DefaultSpoolDirectory” location.



5. Quit Registry Editor.
 6. Restart the Print Spooler service.
 7. All new jobs will be of the form XXXXX.SPL and XXXXX.SHD, and will be numbered consistently with their JobIDs.
- To turn off spool file pooling for all printers:
 1. On the HPAC server, open the Registry Editor.
 2. Locate the following registry subkey:

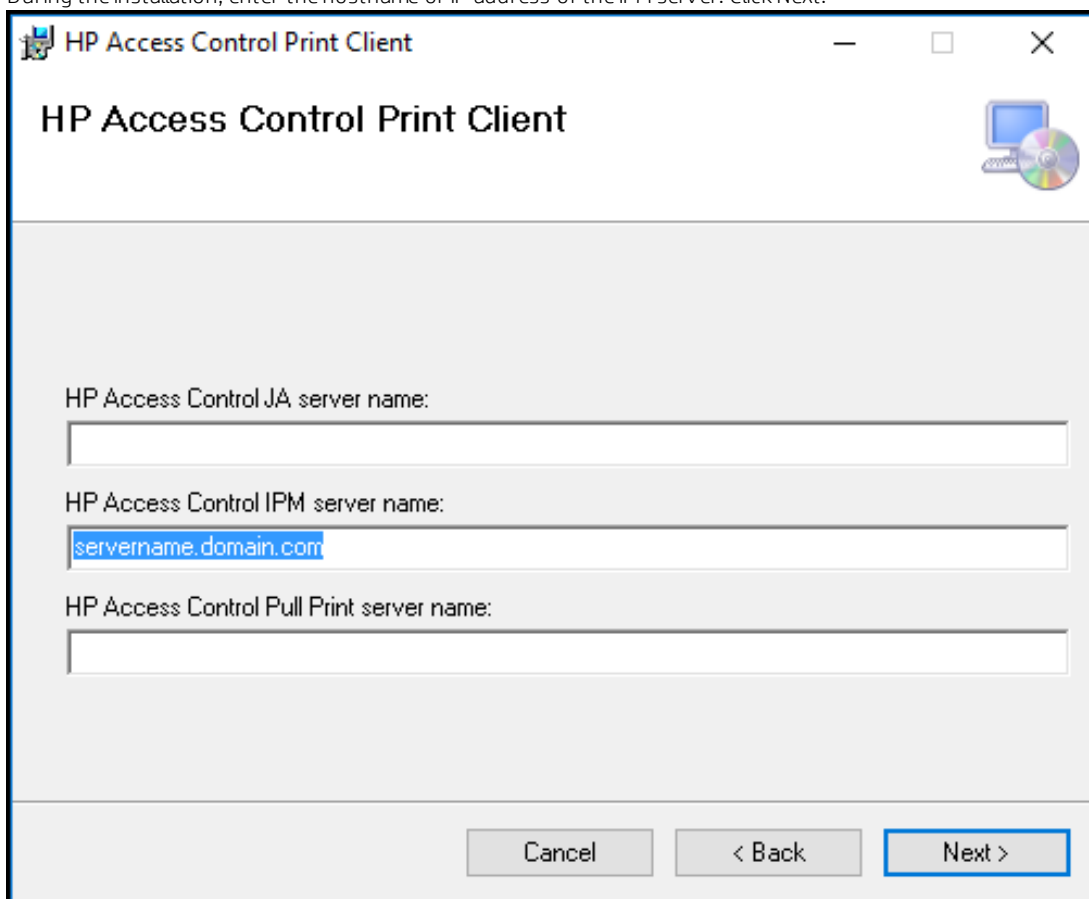
HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\Print
 3. On the Edit menu, point to New, and then click DWORD Value.
 4. In the New Value #1 box, type DisableServerFilePooling, and then press ENTER.
 5. Right-click DisableServerFilePooling, and then click Modify.
 6. In the Value data box, type 1 to turn off spool file pooling, and then click OK.



7. Quit Registry Editor.
8. Restart the Print Spooler service.
9. All new jobs will be of the form XXXXX.SPL and XXXXX.SHD, and will be numbered consistently with their JobIDs.

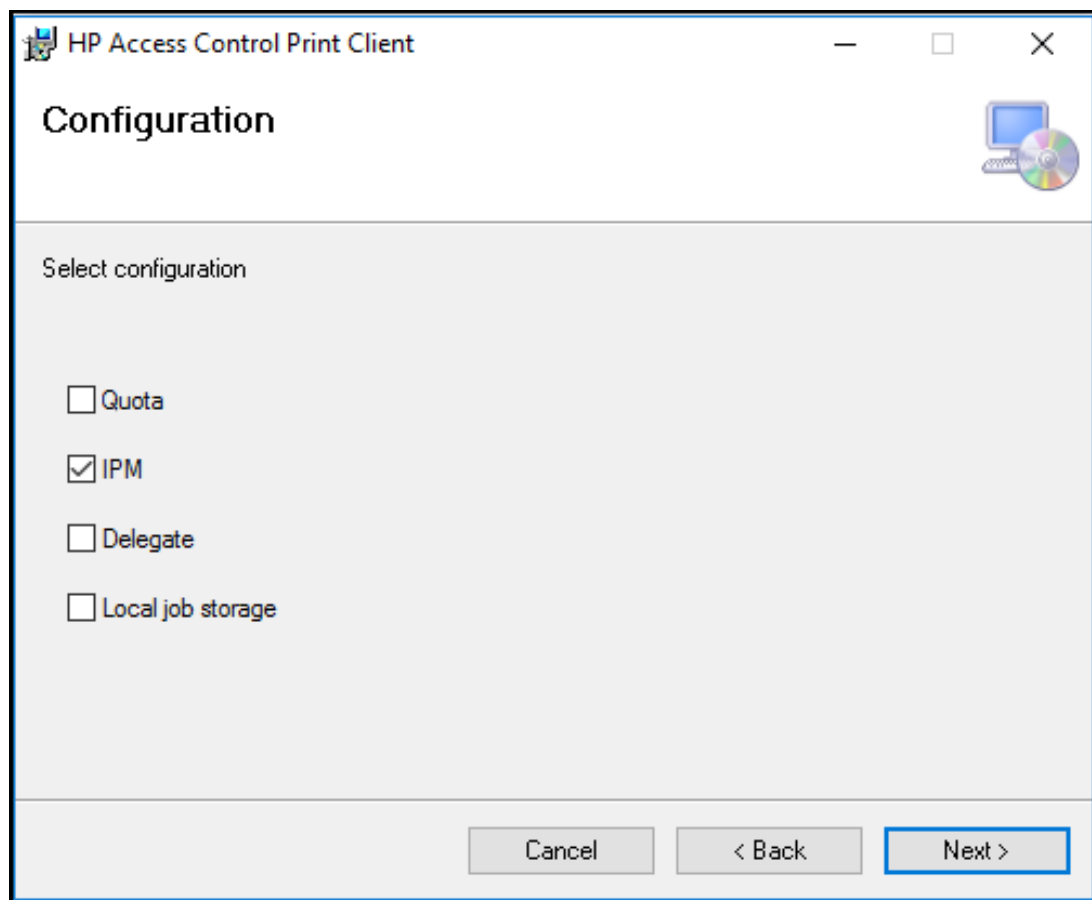
4 Configure pull printing with PDLP

1. On a client workstation, run the HP AC Print Client installer.
2. During the installation, enter the hostname or IP address of the IPM server. Click Next.

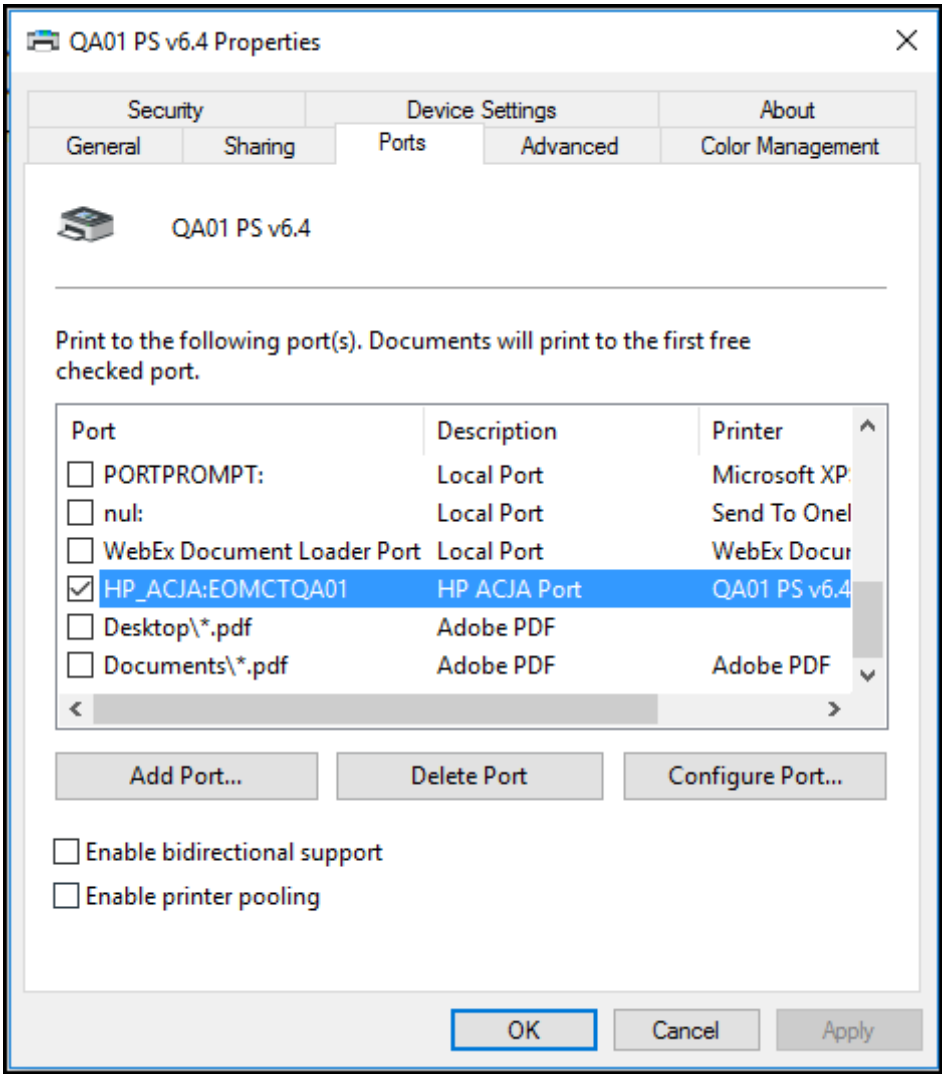


The screenshot shows the 'HP Access Control Print Client' installation window. The title bar includes the HP logo and the text 'HP Access Control Print Client'. The window has standard Windows window controls (minimize, maximize, close) in the top right corner. Below the title bar, there is a sub-header 'HP Access Control Print Client' and a small icon of a computer and a CD. The main area of the window contains three text input fields with labels: 'HP Access Control JA server name:', 'HP Access Control IPM server name:', and 'HP Access Control Pull Print server name:'. The 'HP Access Control IPM server name:' field is currently filled with the text 'servername.domain.com'. At the bottom of the window, there are three buttons: 'Cancel', '< Back', and 'Next >'. The 'Next >' button is highlighted with a blue border.

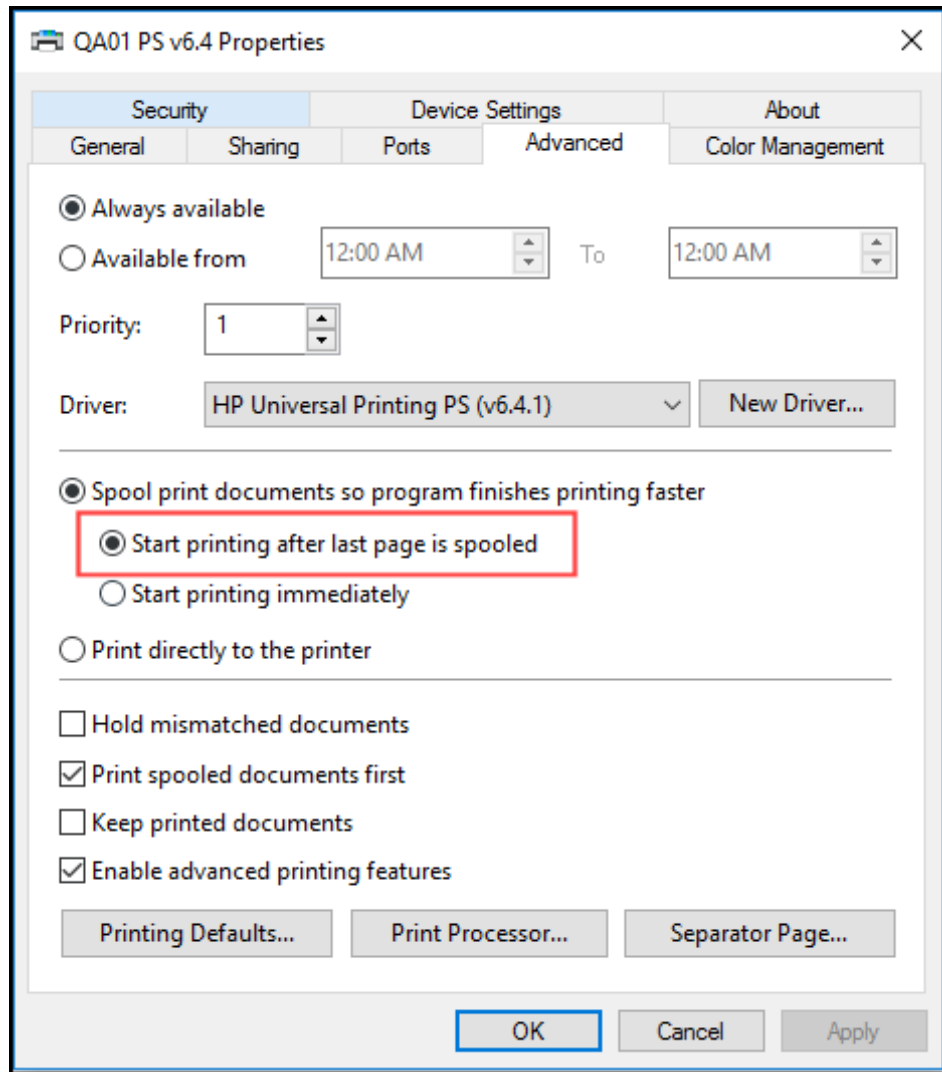
3. Select the IPM checkbox. Click Next.



4. When the installation is complete, click Close.
5. Verify the printer is correctly configured by going to the printer properties > Ports tab, and making sure the port begins with "HP ACJA:".



6. In the printer properties > Advanced tab, make sure “Start printing after last page is spooled” is checked.



7. When users print to this printer, the IPM rule will be triggered.

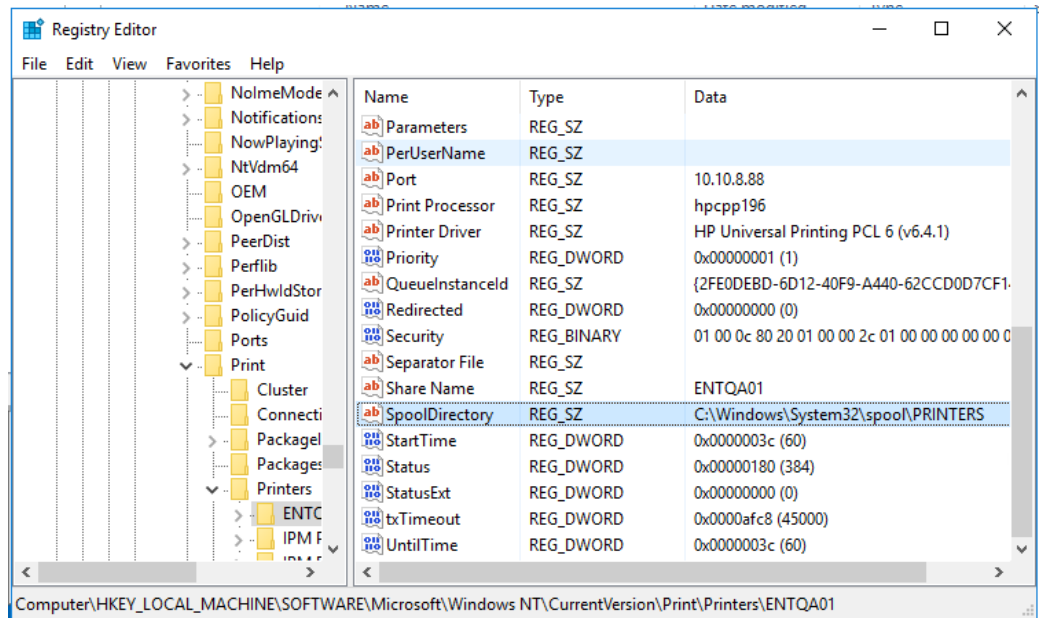
4.1 Turn off spool file pooling

For PDLP to read document content, spool file pooling must be turned off to store spool files as SPL file types. Follow the steps below to turn off spool file pooling per print queue or for all printers:

- To turn off spool file pooling per print queue:
 1. On the client workstation, open the Registry Editor.
 2. Locate the following registry subkey, where <printername> is the name of your printer:
`HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Print\Printers\<printername>`
 3. In the right pane, right-click SpoolDirectory, and then click Modify.

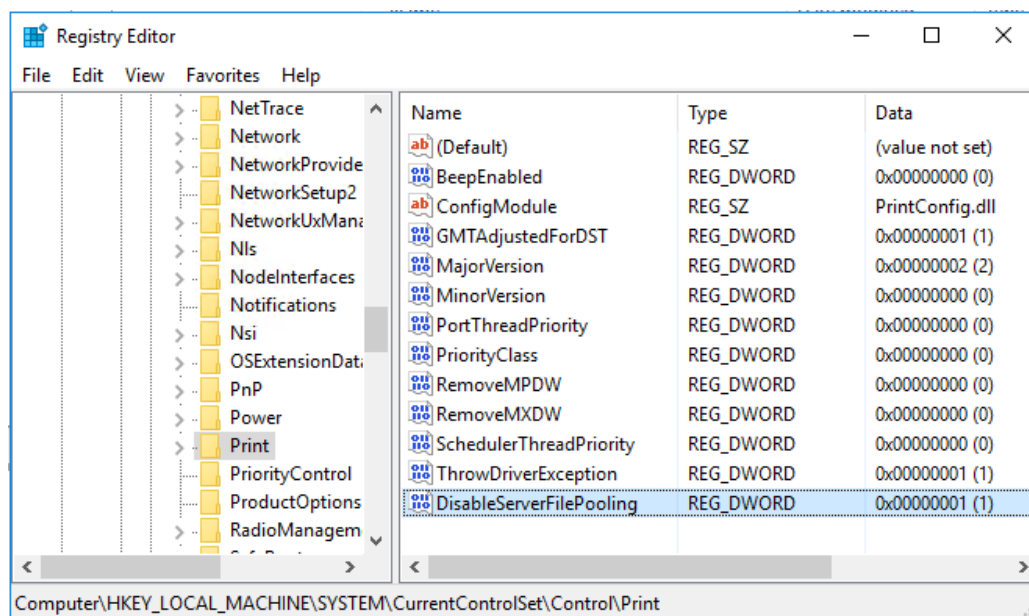
4. In the Value data box, type the location where you want the spool files to be stored. For example, type C:\Windows\System32\spool\PRINTERS as the default location.

IMPORTANT: The “\PRINTERS\<printername>\SpoolDirectory” location must be the same as the “\PRINTERS\DefaultSpoolDirectory” location.



5. Quit Registry Editor.
 6. Restart the Print Spooler service.
 7. All new jobs will be of the form XXXXX.SPL and XXXXX.SHD, and will be numbered consistently with their JobIDs.
- To turn off spool file pooling for all printers:
 1. On the client workstation, open the Registry Editor.
 2. Locate the following registry subkey:

HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\Print
 3. On the Edit menu, point to New, and then click DWORD Value.
 4. In the New Value #1 box, type DisableServerFilePooling, and then press ENTER.
 5. Right-click DisableServerFilePooling, and then click Modify.
 6. In the Value data box, type 1 to turn off spool file pooling, and then click OK.



7. Quit Registry Editor.
8. Restart the Print Spooler service.
9. All new jobs will be of the form XXXXX.SPL and XXXXX.SHD, and will be numbered consistently with their JobIDs.